

Jacob Zheng

☎ +1 613-219-2816 | ✉ jacob.zheng2007@gmail.com | 🔗 linkedin.com/in/jacob-zheng-0069b2304 | 📍 Waterloo, ON

SKILLS

Languages: TypeScript, Java, Python, HTML, CSS, JavaScript, C++, C

Frameworks / Libraries: Node.js, React, Next.js

Developer Tools: Git, GitHub, REST APIs, VS Code

Networking & Backend: HTTP/HTTPS, TCP/IP, Client-Server Architecture

PROJECTS

Tandem (*Next.js, React, TailwindCSS, FastAPI, Vertex AI Gemini, ElevenLabs*)

- Built a real-time, AI study companion that transforms **note-taking into interactive problem solving**.
- Implemented a **conversational tutor** with interruption handling using speech-to-speech pipelines **VAD + ElevenLabs**) for natural back-and-forth dialogue.
- Developed a live whiteboard system **textbf(TLDraw)** enabling the AI to interpret user drawings.
- Engineered a **backend pipeline (FastAPI)** to ingest lecture audio and documents, transcribe and parse content with **Gemini**, and generate structured lesson slides.

Graduation Progress Tracker (*TypeScript, Node.js, React, REST APIs*)

- Built an **academic planning tool** to track degree requirements and student progress.
- Implemented logic to **identify missing credits** and generate recommended next academic steps.
- Designed and tested user workflows with systematic **QA checks**, validating **edge cases** and ensuring **reliable API responses**.

Personal Website (*TypeScript, React, Next.js, HTML, CSS*)

- Designed and developed a **modern, responsive portfolio** with a **clean UI/UX**.
- Optimized the site for performance, readability, and maintainability.

EXPERIENCE

Ski Instructor

Dec 2022 – Jan 2025

Camp Fortune

Chelsea, QC

- Instructed **3 daily lessons** to groups of **10 students**, adapting to varied learning styles while maintaining safety standards and CSIA Level 1 Certification.

Internals Senior Director

Sept 2025 – Present

Vietnamese Students Association (VSA)

Waterloo, ON

- Managed internal operations and communication for a **70-member** organization while coordinating weekly executive meetings for a **20-person leadership team**.

Court Attendant

Sept 2022 – Aug 2025

Barrhaven Tennis Club

Ottawa, ON

- Supported operations for a **250-member** facility, assisting **50+ members per shift** and facilitating bookings, events, and tournaments.

EDUCATION

University of Waterloo

Expected Graduation Apr 2030

Bachelor of Mathematics (Honours)

Waterloo, ON

- Relevant Coursework: CS 135 (Functional Programming), Calculus II, Linear Algebra (Honours Math)
- Clubs: **UW Blueprint**, **WAT.ai**, **Socratica**, **UW Orbit**
- President's Scholarship Recipient

Shad Canada Alumnus

Jun 2024 – Jul 2024

Ontario Tech University

Oshawa, ON

AWARDS AND CERTIFICATIONS

CEMC Euclid Mathematics Contest

— Top 75% Percentile